

Predictive Exit Interview System

→ Explanation:

This project develops a machine learning model that predicts which employees are at a high risk of leaving the company before they actually resign. The predictions help the HR department take preventive actions to improve employee retention.

→ Problem Statement:

Employee attrition is a major challenge for organizations because losing skilled employees increases recruitment costs, training expenses, and decreases overall productivity. The objective of this project is to develop an automated machine learning framework that identifies employees who are at high risk of leaving the organization before they resign. Early identification enables the HR department to take proactive retention measures, improve employee satisfaction, and reduce employee turnover.

→ Business Objective:

The primary objective is to improve employee retention by identifying employees who are likely to leave the organization. The system helps HR professionals make data-driven decisions, reduce recruitment and training costs, improve workforce planning, and increase overall employee satisfaction by taking timely retention actions.

→ Dataset Information

Dataset Name: IBM HR Analytics Employee Attrition Dataset

Source: [Kaggle – IBM HR Analytics Employee Attrition Dataset](#)

Description:

The dataset contains employee-related information such as age, department, salary, job role, overtime, work-life balance, and job satisfaction. It is used to predict whether an employee is likely to leave the organization.

→ Machine Learning Type

Learning Type: Supervised Machine Learning

Problem Type: Binary Classification

Reason:

The dataset contains a labeled target variable (Attrition) with two possible outcomes:

- **Yes** – Employee is likely to leave the company.
- **No** – Employee is likely to stay with the company.

The model learns from historical employee data to predict the attrition status of new employees.

→ **Input Features**

The model uses the following employee attributes as input features:

- Age
- BusinessTravel
- DailyRate
- Department
- DistanceFromHome
- Education
- EducationField
- EmployeeCount
- EmployeeNumber
- EnvironmentSatisfaction
- Gender
- HourlyRate
- JobInvolvement
- JobLevel
- JobRole
- JobSatisfaction
- MaritalStatus
- MonthlyIncome
- MonthlyRate
- NumCompaniesWorked
- Over18
- OverTime
- PercentSalaryHike
- PerformanceRating
- RelationshipSatisfaction
- StandardHours
- StockOptionLevel
- TotalWorkingYears
- TrainingTimesLastYear
- WorkLifeBalance
- YearsAtCompany
- YearsInCurrentRole
- YearsSinceLastPromotion
- YearsWithCurrManager

Note: During preprocessing, columns such as EmployeeCount, EmployeeNumber, Over18, and StandardHours are commonly removed because they are identifiers or constant-value features that do not contribute meaningfully to prediction.

→ **Target Feature: Attrition**

Type: Binary Categorical Variable

Possible Values:

- **Yes (1):** Employee is likely to leave the organization.
- **No (0):** Employee is likely to remain with the organization.

Purpose:

The machine learning model predicts the value of the Attrition column based on the employee's characteristics. Employees predicted as "Yes" are considered high-risk attrition candidates, allowing the HR department to take proactive retention measures before they resign.

→ **Project Scope**

The scope of this project is to develop a machine learning-based predictive system that identifies employees who are at risk of leaving the organization. The project involves collecting the IBM HR Analytics Employee Attrition dataset, preprocessing the data, encoding categorical variables, selecting relevant features, splitting the data into training and testing sets, and training a classification model (Random Forest Classifier). The model will be evaluated using standard performance metrics, and feature importance analysis will be performed to identify the factors that most influence employee attrition. Finally, the system will predict high-risk employees and provide insights that can help the HR department implement effective employee retention strategies.

→ **Evaluation Metrics**

The performance of the machine learning model will be evaluated using the following metrics:

- **Accuracy:** Measures the overall percentage of correctly classified employees.
- **Precision:** Measures how many employees predicted to leave actually leave the organization.
- **Recall (Sensitivity):** Measures how effectively the model identifies employees who are likely to leave.
- **F1-Score:** Provides a balanced measure of precision and recall, especially useful for imbalanced datasets.
- **Confusion Matrix:** Displays the number of correct and incorrect predictions for each class.
- **ROC-AUC Score (Optional):** Evaluates the model's ability to distinguish between employees who stay and those who leave.

→ **Expected Deliverables**

The following deliverables will be produced upon completion of the project:

1. **ML Project Charter (PDF)** describing the project objectives, scope, dataset, and methodology.
2. **Feature Importance Table** showing the most influential factors affecting employee attrition.
3. **Workflow Pipeline Diagram** illustrating the complete machine learning process from data collection to prediction.
4. **Jupyter Notebook (.ipynb)** containing data preprocessing, model development, evaluation, and prediction.
5. **Predictive Model** capable of identifying employees at high risk of attrition.
6. **Prediction Results** highlighting employees classified as high-risk.

→ **Expected Benefits**

The proposed Predictive Exit Interview System is expected to provide the following benefits:

- Early identification of employees who are at high risk of leaving the organization.
- Improved employee retention through timely HR interventions.
- Reduced recruitment, onboarding, and training costs associated with employee turnover.
- Better workforce planning and resource management.
- Data-driven decision-making for HR professionals.
- Identification of key factors contributing to employee attrition, enabling targeted improvement strategies.
- Increased employee satisfaction and organizational productivity through proactive retention efforts.